

torch.complex#

<https://pytorch.org/docs/stable/generated/torch.complex.html>

Description:

Constructs a complex tensor with its real part equal to real and its imaginary part equal to imag. real (Tensor) – The real part of the complex tensor. Must be half, float or double. imag (Tensor) – The imaginary part of the complex tensor. Must be same dtype as real. out (Tensor) – If the inputs are torch.float32, must be torch.complex64. If the inputs are torch.float64, must be torch.complex128.

Function Signature

```
torch.complex(real, imag, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor) – If the inputs are torch.float32, must be torch.complex64. If the inputs are torch.float64, must be torch.complex128.

Code Examples

```
>>> real = torch.tensor([1, 2], dtype=torch.float32)
>>> imag = torch.tensor([3, 4], dtype=torch.float32)
>>> z = torch.complex(real, imag)
>>> z
tensor([(1.+3.j), (2.+4.j)])
>>> z.dtype
torch.complex64
```

Detailed Documentation

Documentation

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`real` (Tensor) – The real part of the complex tensor. Must be half, float or double.

`imag` (Tensor) – The imaginary part of the complex tensor. Must be same dtype as `real`.

`out` (Tensor) – If the inputs are `torch.float32`, must be `torch.complex64`. If the inputs are `torch.float64`, must be `torch.complex128`.

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